DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
VNR VJIET

Week No: 9	**Date:**	**Roll No**: 23071A05H0
Experiment: XML structure designed to store user account details including username, encrypted password, email, and purchase history.		

Aim :

To develop a web-based user authentication system using **Java Servlets and JDBC**, enabling user registration, login, session management, and logout functionality.

Description :

This experiment implements a basic authentication system where users can register and log in using their credentials stored in a database. Java Servlets handle client requests and responses, while JDBC is used to interact with the database for storing and validating user data. Session tracking using HttpSession ensures that only authenticated users can access protected pages. The system demonstrates the integration of frontend forms with backend processing and database connectivity.

Concepts Used :

Java Servlet Concepts

- HttpServlet – Base class for creating servlets
- doPost() / doGet() – Handles client requests
- HttpServletRequest – Retrieves form data
- HttpServletResponse – Sends response to browser
- HttpSession – Maintains user login state
- RequestDispatcher – Forwards request to JSP pages
- sendRedirect() – Redirects user after login/logout

JDBC Concepts

- DriverManager – Establishes DB connection
- Connection – Connects to database
- PreparedStatement – Executes secure SQL queries
- ResultSet – Stores query results
- SQL Operations – INSERT (register), SELECT (login validation)

Steps to Execute :

Start

↓

Create Database (wt_lab)

↓

Create users table

↓

Insert sample data (optional)

↓

Create Dynamic Web Project in Eclipse

↓

Configure Apache Tomcat Server

↓

Add MySQL Connector (JAR file)

↓

Create Servlets (Register, Login, Logout)

↓

Write JDBC code (Connect → Insert → Validate)

↓

Deploy project on Tomcat

↓

Open browser (register.jsp → login.jsp)

↓

Perform registration and login

↓

Access home page (session-based)

↓

Logout and invalidate session

↓

End

Code Implementation :

1. DBUtil.java

```
import java.sql.Connection;
import java.sql.DriverManager;
public class DBUtil {
    private static final String URL  = "jdbc:mysql://localhost:3306/wt_lab";
    private static final String USER = "root";
    private static final String PASS = "your_password";
    public static Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return DriverManager.getConnection(URL, USER, PASS);
    }
}
```

2. LoginServlet.java

```
import java.io.IOException;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;

import javax.servlet.*;
import javax.servlet.http.*;

public class LoginServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        String email = request.getParameter("email");
        String password = request.getParameter("password");

        String sql = "SELECT email FROM users WHERE email=? AND password=?";

        try (Connection con = DBUtil.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setString(1, email);
            ps.setString(2, password);

            ResultSet rs = ps.executeQuery();
```

```
            if (rs.next()) {
                HttpSession session = request.getSession();
                session.setAttribute("user", email);
                response.sendRedirect("home.jsp");
            } else {
                request.setAttribute("error", "Invalid Credentials - 23071A05H0");
                request.getRequestDispatcher("login.jsp").forward(request, response);
            }

        } catch (Exception e) {
            request.setAttribute("error", "Database Error - 23071A05H0");
            request.getRequestDispatcher("login.jsp").forward(request, response);
        }
    }
}
```

3. RegisterServlet.java

```
import java.io.IOException;
import java.sql.Connection;
import java.sql.PreparedStatement;

import javax.servlet.*;
import javax.servlet.http.*;

public class RegisterServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        String email = request.getParameter("email");
        String password = request.getParameter("password");

        String sql = "INSERT INTO users(email, password) VALUES(?, ?)";

        try (Connection con = DBUtil.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setString(1, email);
            ps.setString(2, password);

            int rows = ps.executeUpdate();

            if (rows > 0) {
                request.setAttribute("msg", "Registration Successful! - 23071A05H0");
                request.getRequestDispatcher("login.jsp").forward(request, response);
```

```
            } else {
                request.setAttribute("msg", "Registration Failed - 23071A05H0");
                request.getRequestDispatcher("errorLogin.jsp").forward(request, response);
            }

        } catch (Exception e) {
            request.setAttribute("msg", "User already exists / DB Error - 23071A05H0");
            request.getRequestDispatcher("errorLogin.jsp").forward(request, response);
        }
    }
}
```

4. LogoutServlet.java

```
import java.io.IOException;
import javax.servlet.*;
import javax.servlet.http.*;

public class LogoutServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        HttpSession session = request.getSession(false);

        if (session != null) {
            session.invalidate();
        }

        response.sendRedirect("login.jsp");
    }
}
```

5. style.css

```
body {
    background: linear-gradient(135deg, #667eea, #764ba2);
    font-family: Arial;
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
}
```

```
.container {
    background: white;
    padding: 30px;
    border-radius: 10px;
    width: 350px;
    text-align: center;
}

input {
    width: 100%;
    padding: 10px;
    margin: 10px 0;
}
.error { color: red; }
.success { color: green; }
```

6. web.xml

```
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee" version="4.0">
  <servlet>
    <servlet-name>LoginServlet</servlet-name>
    <servlet-class>LoginServlet</servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name>LoginServlet</servlet-name>
    <url-pattern>/login</url-pattern>
  </servlet-mapping>

  <servlet>
    <servlet-name>RegisterServlet</servlet-name>
    <servlet-class>RegisterServlet</servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name>RegisterServlet</servlet-name>
    <url-pattern>/register</url-pattern>
  </servlet-mapping>

  <servlet>
    <servlet-name>LogoutServlet</servlet-name>
    <servlet-class>LogoutServlet</servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name>LogoutServlet</servlet-name>
    <url-pattern>/logout</url-pattern>
  </servlet-mapping>

</web-app>
```

7. home.jsp

```
<%
if(session == null || session.getAttribute("user") == null){
    request.setAttribute("msg","Please login first to access this page");
    request.getRequestDispatcher("errorLogin.jsp").forward(request,response);
    return;
}
String user=(String)session.getAttribute("user");
%>

<!DOCTYPE html>
<html>
<head>
<title>Home</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body>

<div class="container">
<h2>Welcome <%= user %></h2>
<h3>Project: 23071A05H0 WEEK9</h3>

<a href="logout">Logout</a>
</div>

</body>
</html>
```

8. login.jsp

```
<!DOCTYPE html>
<html>
<head>
<title>Login</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body>
<div class="container">
<h2>Login</h2>

<form action="login" method="post">
    <input type="text" name="email" placeholder="Enter Email" required>
    <input type="password" name="password" placeholder="Enter Password" required>
    <input type="submit" value="Login">
</form>
```

```
<p class="error">
<%= request.getAttribute("error") != null ? request.getAttribute("error") : "" %>
</p>

<a href="register.jsp">Register</a>
</div>

</body>
</html>
```

9. register.jsp

```
<!DOCTYPE html>
<html>
<head>
<title>Register</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body>

<div class="container">
<h2>Register</h2>

<form action="register" method="post">
   <input type="text" name="email" placeholder="Enter Email" required>
   <input type="password" name="password" placeholder="Enter Password" required>
   <input type="submit" value="Register">
</form>

<p class="success">
<%= request.getAttribute("msg") != null ? request.getAttribute("msg") : "" %>
</p>

<a href="login.jsp">Back to Login</a>
</div>

</body>
</html>
```

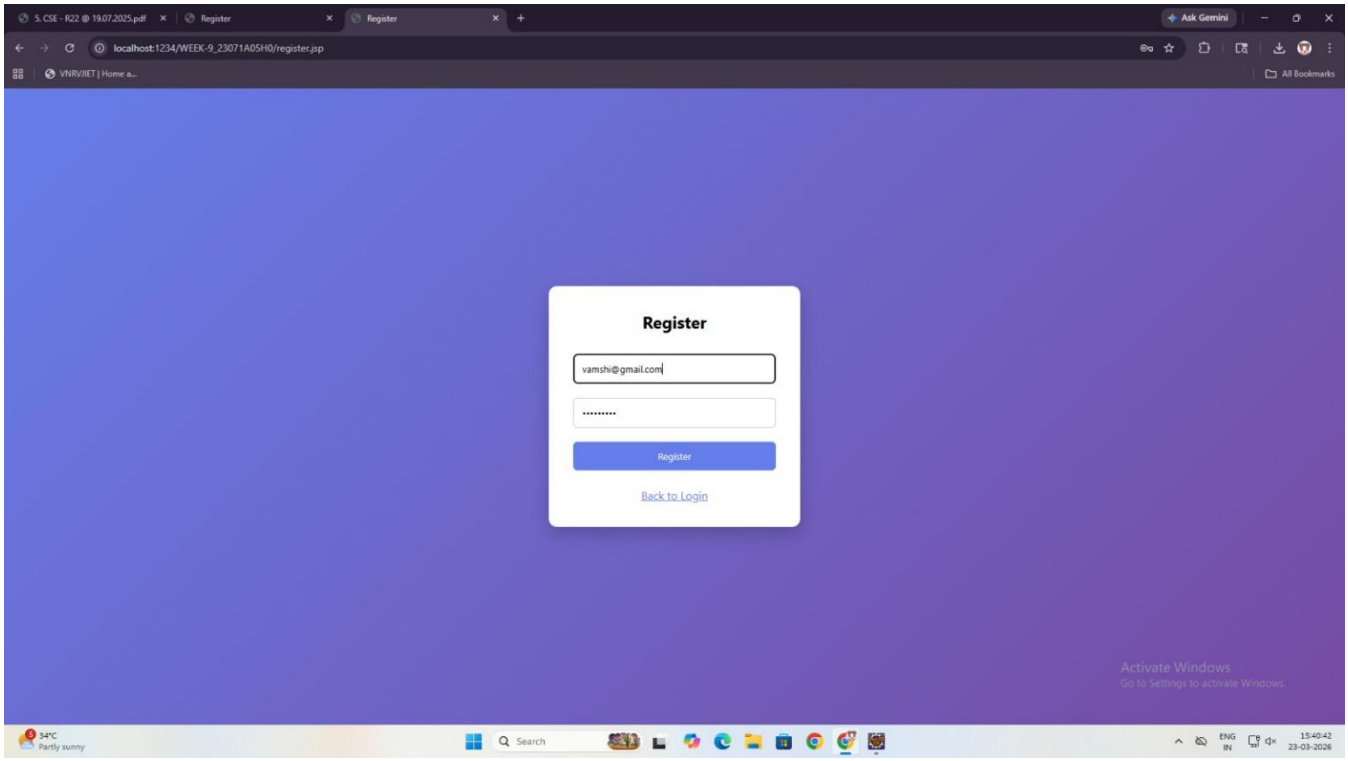
10. error.jsp

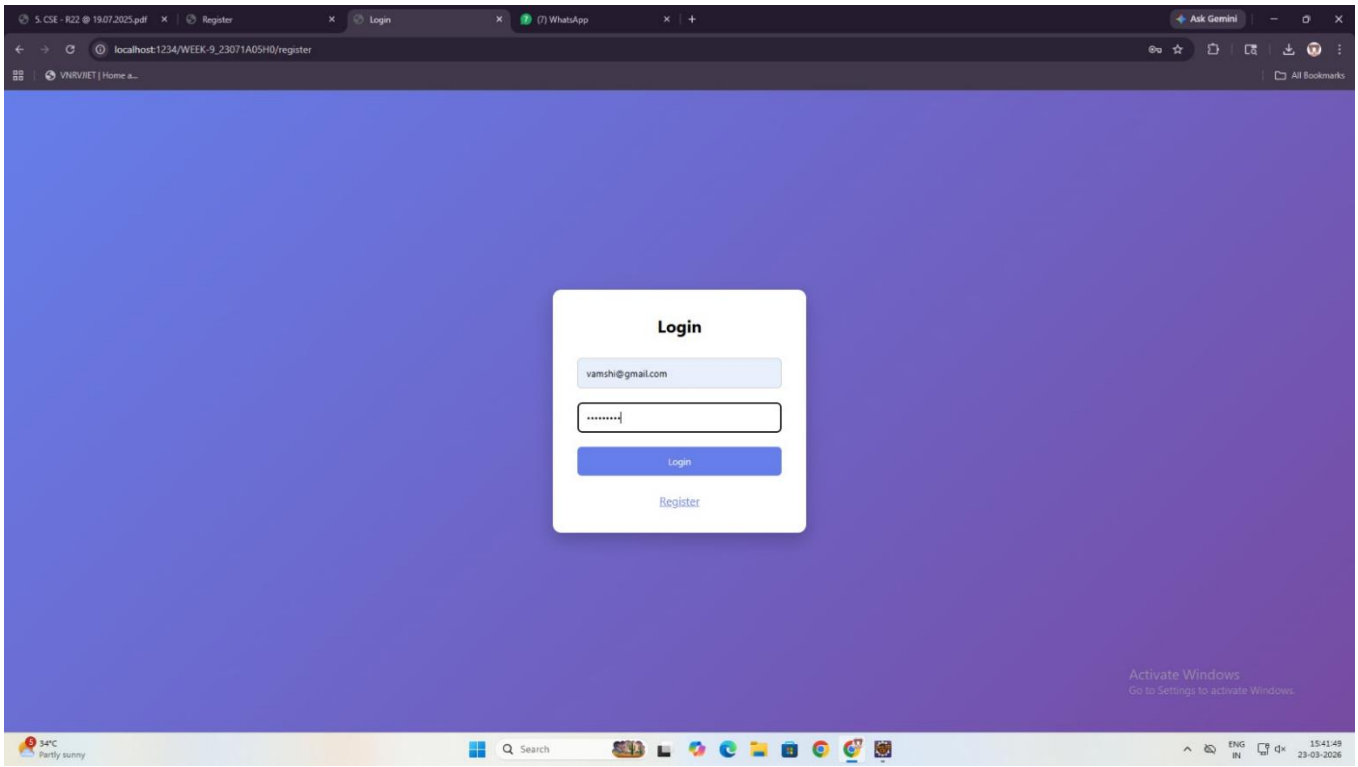
```
<body>
<div class="container">
  <h3 class="error">Invalid Login - 23071A05H0</h3>
  <a href="login.jsp">Try Again</a>
</div>
</body>
```

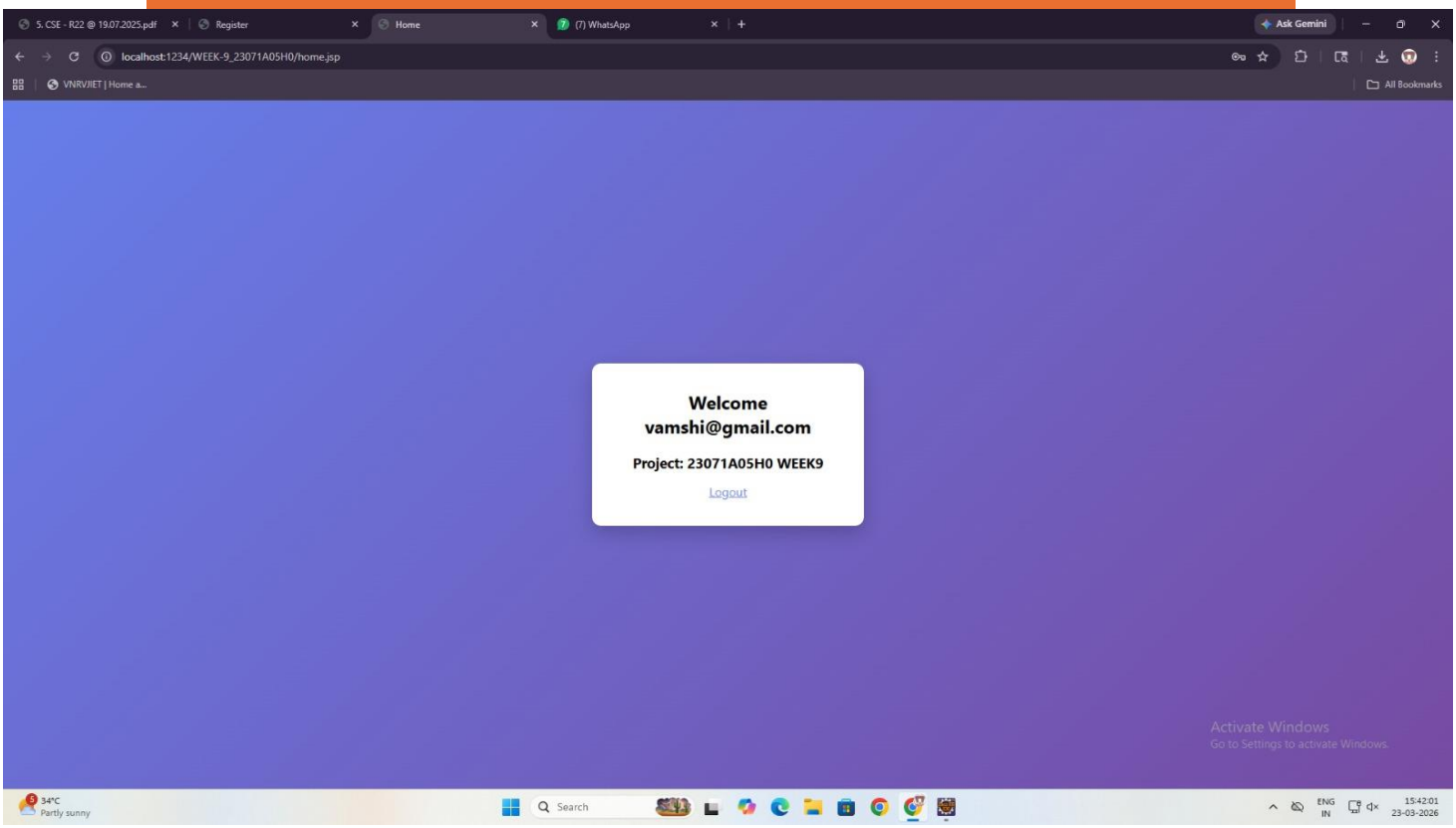
11. errorLogin.jsp

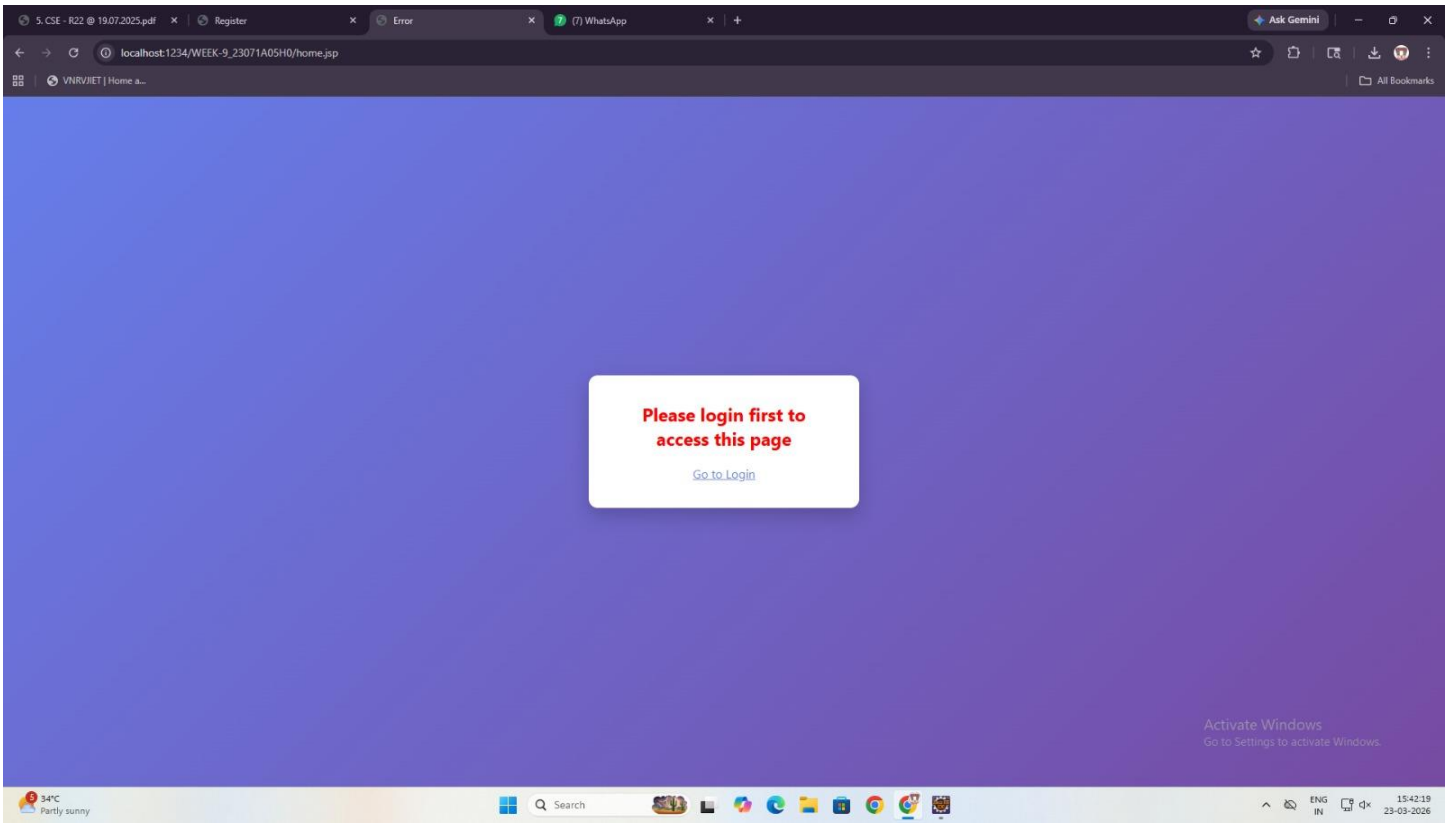
```
<!DOCTYPE html>
<html>
<head>
<title>Error</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body>

<div class="container">
<h2 class="error">
<%= request.getAttribute("msg") != null ? request.getAttribute("msg") : "Access Denied" %>
</h2>

<a href="login.jsp">Go to Login</a>
</div>

</body>
</html>
```

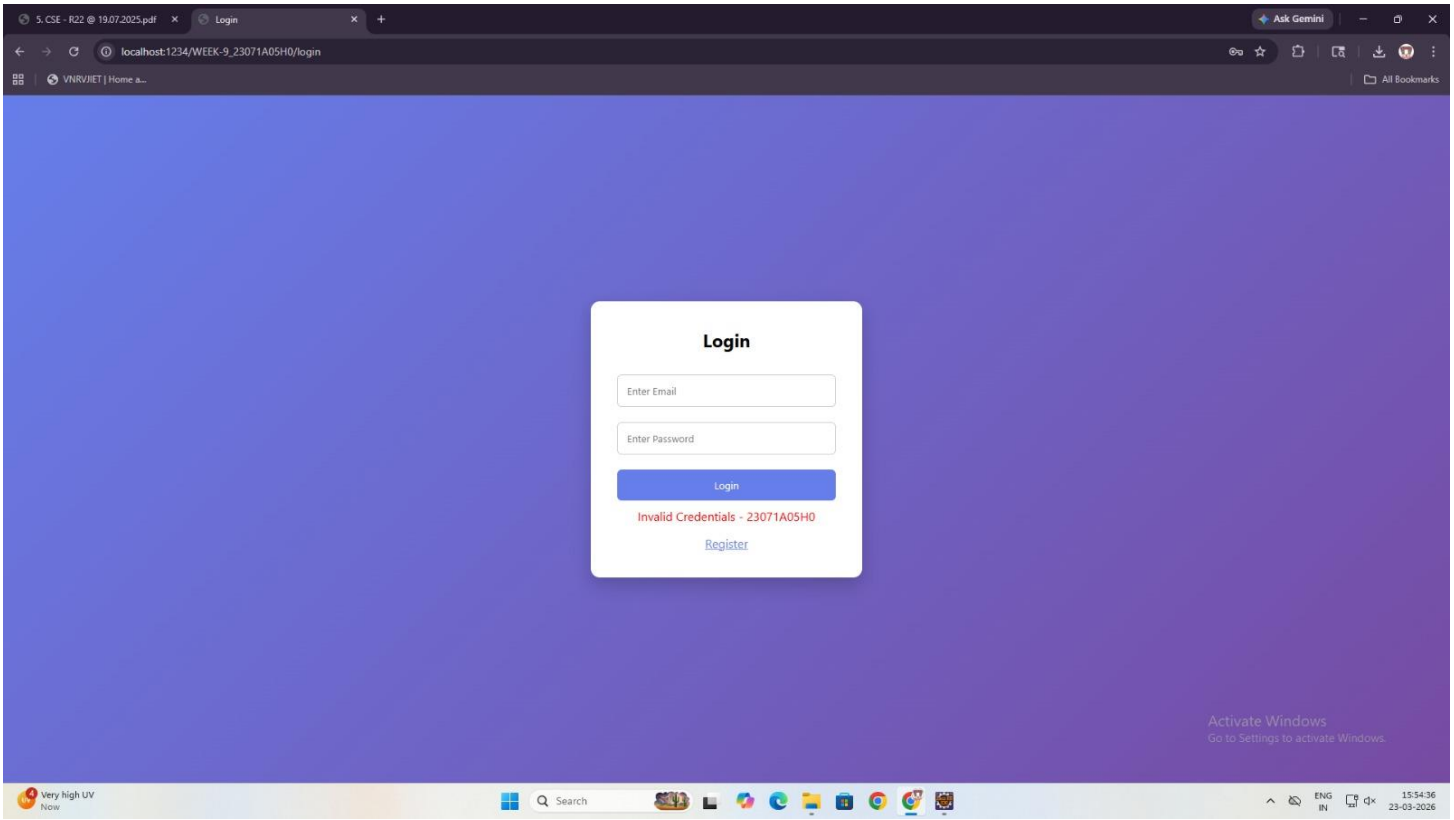
OUTPUT :


Register
vamshi@gmail.com
Register
Back to Login


Login
vamshi@gmail.com
Login
Register


5. CSE - R22 @ 19.07.2025.pdf
Register
Home
(7) WhatsApp
Ask Gemini
localhost:1234/WEEK-9_23071A05H0/home.jsp
VNRVJIET | Home a...
All Bookmarks
Welcome
vamshi@gmail.com
Project: 23071A05H0 WEEK9
Logout
Activate Windows
Go to Settings to activate Windows.
34°C
Partly sunny
Search
ENG
IN
15:42:01
23-03-2026


5. CSE - R22 @ 19.07.2025.pdf
Register
Error
(7) WhatsApp
Ask Gemini
localhost:1234/WEEK-9_23071A05H0/home.jsp
VNRVJIET | Home a...
All Bookmarks
Please login first to access this page
Go to Login
Activate Windows
Go to Settings to activate Windows.
34°C
Partly sunny
Search
ENG
IN
15:42:19
23-03-2026


5. CSE - R22 @ 19.07.2025.pdf
Login
Ask Gemini
localhost:1234/WEEK-9_23071A05H0/login
VNRVJIET | Home a...
All Bookmarks
Login
Enter Email
Enter Password
Login
Invalid Credentials - 23071A05H0
Register
Activate Windows
Go to Settings to activate Windows.
Very high UV
Now
Search
ENG
IN
15:54:36
23-03-2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
VNR VJIET

Week No: 13	**Date:**	**Roll No**: 23071A05H0
Experiment: To implement a checkout process using Servlets and JSP that allows users to review their cart, enter shipping details, and complete the purchase.		

Aim :
To implement a checkout process using Servlets and JSP that allows users to review their cart, enter shipping details, and complete the purchase.

Description :
In this experiment, a complete checkout workflow is developed for an online store using JSP and Servlets.

Users can browse products, add them to a shopping cart, and modify quantities. The cart data is stored in the session using Java collections.

During checkout, users review their selected items, view the total cost, and provide shipping information such as name and address. The CheckoutServlet processes the order, calculates the final amount, and confirms the purchase.

This experiment demonstrates session management, dynamic cart handling, and real-time order processing in a web application.

Technologies Used:

- JSP – User interface
- Servlets – Business logic
- HTTP Session – Store cart data
- Java Collections (Map) – Manage cart items
- HTML/CSS – Page structure and styling

Concepts Used:

- Session tracking (HttpSession)
- Form handling (request.getParameter)
- Cart management using HashMap
- Dynamic content rendering in JSP
- Total price calculation
- Request redirection (sendRedirect)

Steps to Execute:

Start

↓

Create JSP pages → login.jsp, register.jsp, catalog.jsp, cart.jsp, checkout.jsp

↓

Create Servlets → LoginServlet, RegisterServlet, CartServlet, CheckoutServlet

↓

Deploy project on Tomcat

↓

Register user

↓

Login to system

↓

Browse catalog → Add items to cart

↓

View cart → Modify quantity

↓

Proceed to checkout

↓

Enter shipping details

↓

Place order → Order confirmation displayed

↓

End

Code Implementation:

1. catalog.jsp

```
<html>
<head>
<title>Catalog</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body>

<h1>Product Catalog</h1>

<form action="CartServlet" method="post">
   <h3>Laptop</h3>
   <p>Price: 60000</p>
   <input type="hidden" name="productId" value="Laptop">
   <input type="hidden" name="price" value="60000">
   <button name="action" value="add">Add to Cart</button>
</form>

<form action="CartServlet" method="post">
   <h3>Mobile</h3>
   <p>Price: 25000</p>
   <input type="hidden" name="productId" value="Mobile">
   <input type="hidden" name="price" value="25000">
   <button name="action" value="add">Add to Cart</button>
</form>

<form action="CartServlet" method="post">
   <h3>Headphones</h3>
   <p>Price: 3000</p>
   <input type="hidden" name="productId" value="Headphones">
   <input type="hidden" name="price" value="3000">
   <button name="action" value="add">Add to Cart</button>
</form>

<br>
<a href="cart.jsp">View Cart</a>

</body>
</html>
```

2. CartServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import java.util.*;

public class CartServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
            throws ServletException, IOException {

        String product = request.getParameter("productId");
        String action = request.getParameter("action");
        int price = Integer.parseInt(request.getParameter("price"));

        HttpSession session = request.getSession();

        Map<String, Integer> cart =
                (Map<String, Integer>) session.getAttribute("cart");

        Map<String, Integer> prices =
                (Map<String, Integer>) session.getAttribute("prices");

        if (cart == null) {
            cart = new HashMap<>();
            prices = new HashMap<>();
        }

        int qty = cart.getOrDefault(product, 0);
        if ("add".equals(action)) {
            cart.put(product, qty + 1);
            prices.put(product, price);
        } else if ("remove".equals(action)) {
            if (qty > 1) {
                cart.put(product, qty - 1);
            } else {
                cart.remove(product);
                prices.remove(product);
            }
        }
        session.setAttribute("cart", cart);
        session.setAttribute("prices", prices);
        response.sendRedirect("cart.jsp");
    }
}
```

3. cart.jsp

```
<%@ page import="java.util.*" %>
<%
Map<String,Integer> cart = (Map<String,Integer>) session.getAttribute("cart");
Map<String,Integer> prices = (Map<String,Integer>) session.getAttribute("prices");
%>

<html>
<head>
<link rel="stylesheet" href="css/style.css">
</head>
<body>

<h1>Your Cart</h1>
<%
if(cart == null || cart.isEmpty()){
%>
<p>Cart is empty</p>
<%
} else {
   for(String item : cart.keySet()){
%>

<form action="CartServlet" method="post">
   <p>
   <%= item %> - Qty: <%= cart.get(item) %> - Rs.<%= prices.get(item) %>
   </p>

   <input type="hidden" name="productId" value="<%= item %>">
   <input type="hidden" name="price" value="<%= prices.get(item) %>">

   <button name="action" value="add">+</button>
   <button name="action" value="remove">-</button>
</form>

<%
   }
}
%>
<br>
<a href="catalog.jsp">Continue Shopping</a>
<br><br>
<a href="checkout.jsp">Proceed to Checkout</a>

</body>
</html>
```

4. checkout.jsp

```
<%@ page import="java.util.*" %>

<%
Map<String,Integer> cart = (Map<String,Integer>) session.getAttribute("cart");
Map<String,Integer> prices = (Map<String,Integer>) session.getAttribute("prices");

int total = 0;
%>

<html>
<head>
<link rel="stylesheet" href="css/style.css">
</head>
<body>

<h1>Checkout</h1>

<%
if(cart != null){
    for(String item : cart.keySet()){
        int sub = cart.get(item) * prices.get(item);
        total += sub;
%>
<p><%= item %> - Rs.<%= sub %></p>
<%
    }
}
%>

<h3>Total: Rs.<%= total %></h3>

<form action="CheckoutServlet" method="post">
    Name: <input type="text" name="name"><br><br>
    Address: <input type="text" name="address"><br><br>
    <button type="submit">Place Order</button>
</form>

</body>
</html>
```

5. CheckoutServlet.java

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;
import java.util.*;

public class CheckoutServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
            throws ServletException, IOException {

        HttpSession session = request.getSession();

        Map<String,Integer> cart =
                (Map<String,Integer>) session.getAttribute("cart");

        Map<String,Integer> prices =
                (Map<String,Integer>) session.getAttribute("prices");

        int total = 0;

        if(cart != null){
            for(String item : cart.keySet()){
                total += cart.get(item) * prices.get(item);
            }
        }

        session.removeAttribute("cart");
        session.removeAttribute("prices");

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<h2>Order Successful</h2>");
        out.println("<p>Total Paid: Rs." + total + "</p>");
        out.println("<a href='catalog.jsp'>Continue Shopping</a>");
    }
}
```

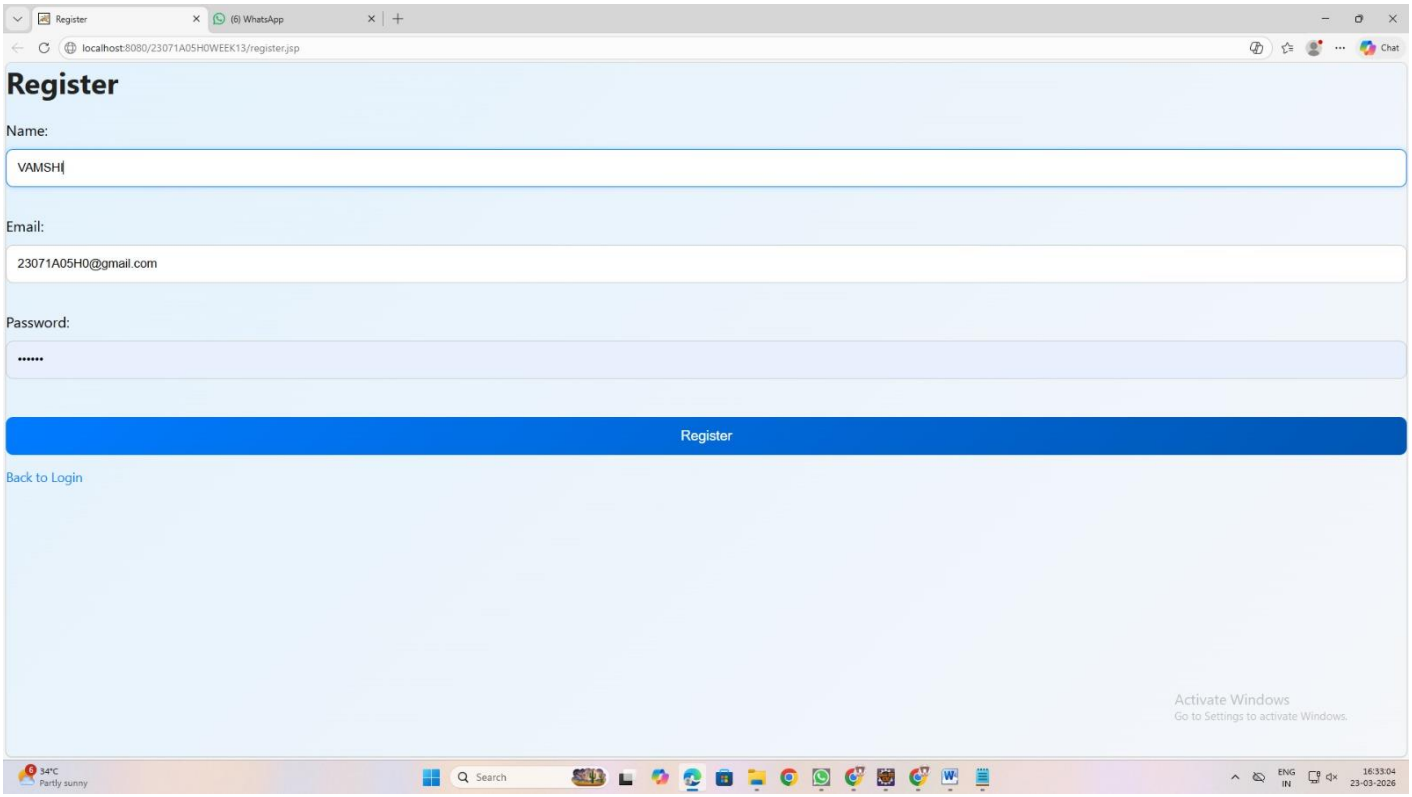
6. LoginServlet.java

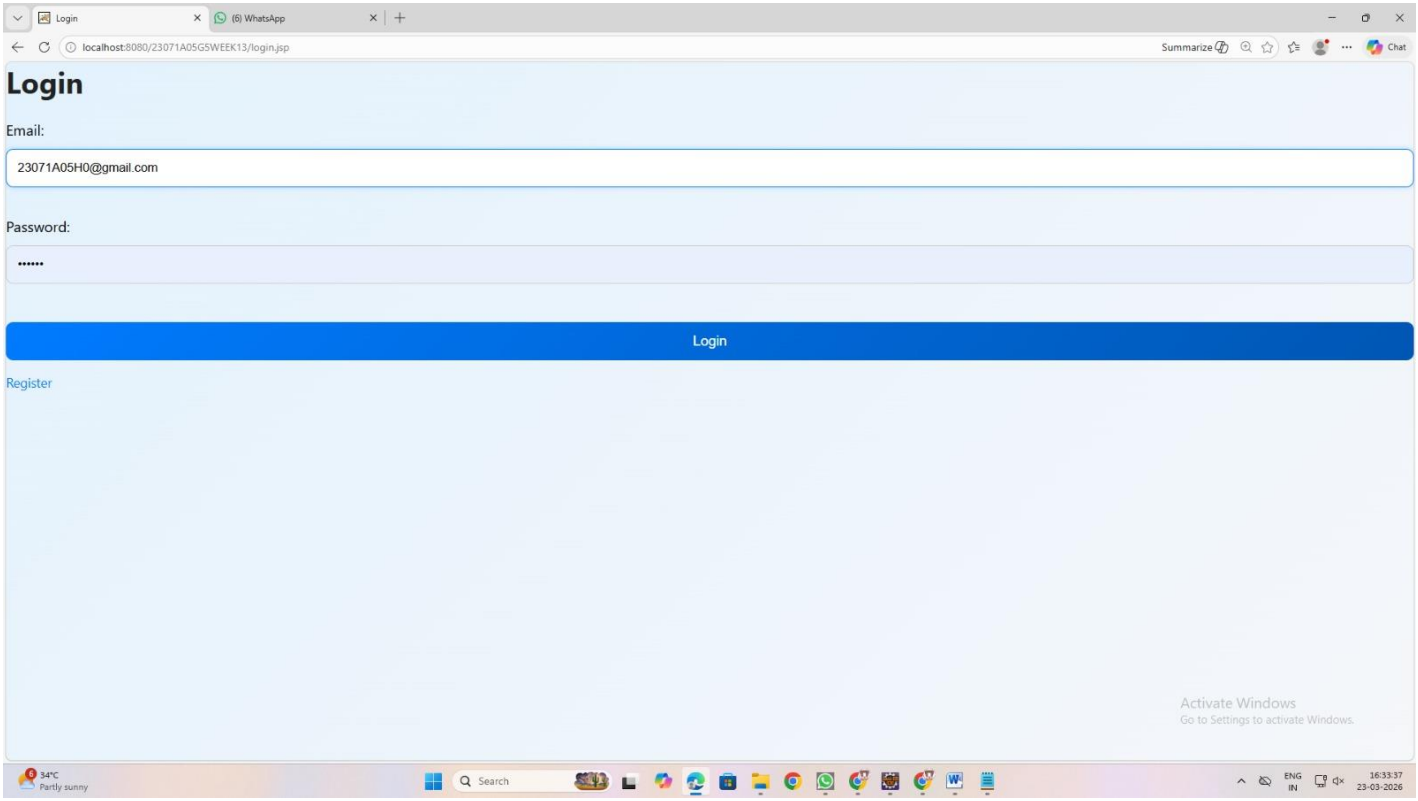
```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;
public class LoginServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
        throws ServletException, IOException {

        String email = request.getParameter("email");
        String password = request.getParameter("password"); // ❗ FIXED missing ;

        HttpSession session = request.getSession();

        String savedEmail = (String) session.getAttribute("userEmail");
        String savedPassword = (String) session.getAttribute("userPassword");

        if(email.equals(savedEmail) && password.equals(savedPassword)){
            response.sendRedirect("catalog.jsp");
        } else {
            response.getWriter().println("Invalid Login");
        }
    }
}
```

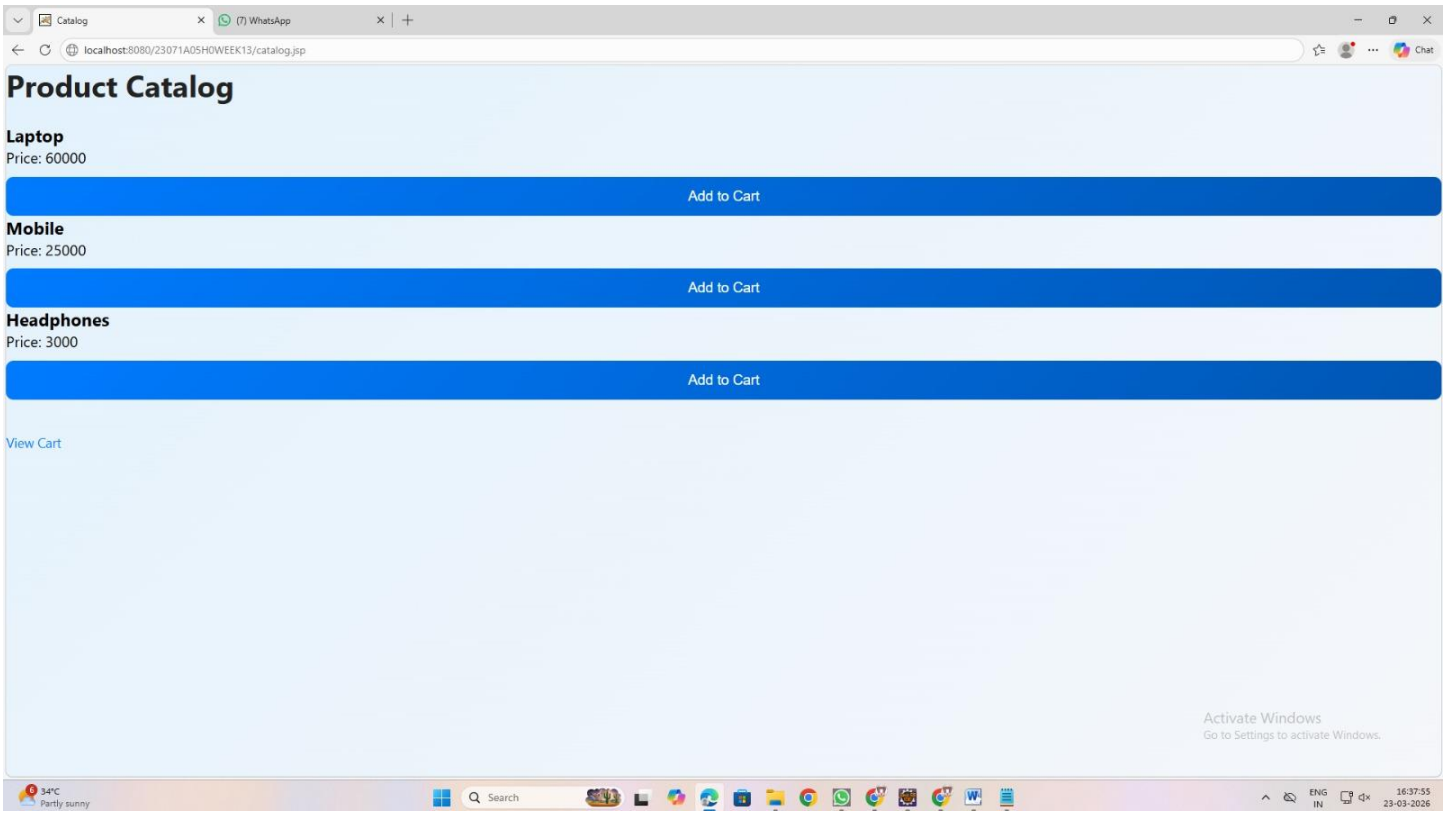
7. RegisterServlet.java

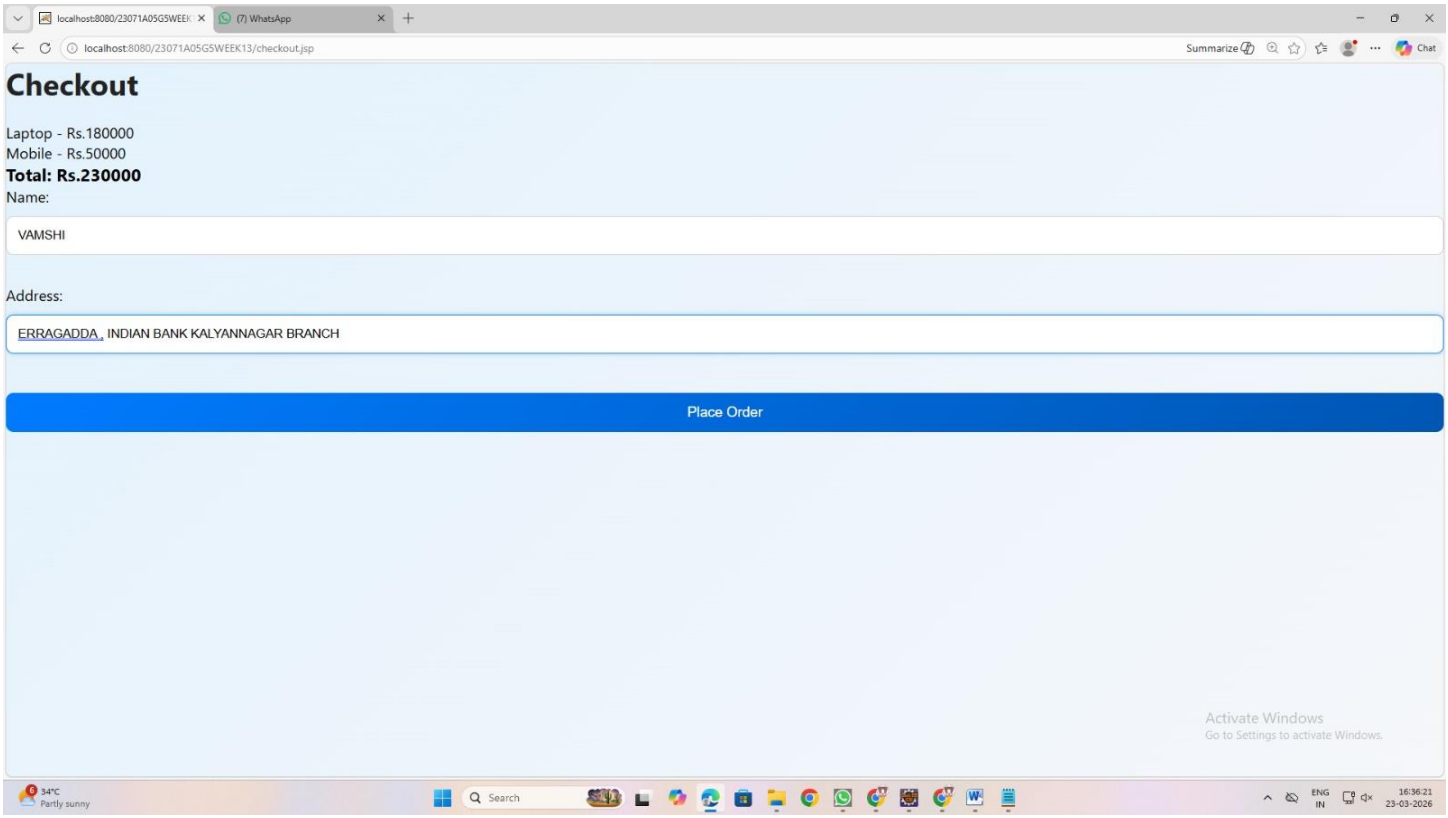
```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;
public class RegisterServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
        throws ServletException, IOException {

        String email = request.getParameter("email");
        String password = request.getParameter("password");

        HttpSession session = request.getSession();
        session.setAttribute("userEmail", email);
        session.setAttribute("userPassword", password);
        response.sendRedirect("login.jsp");
    }
}
```

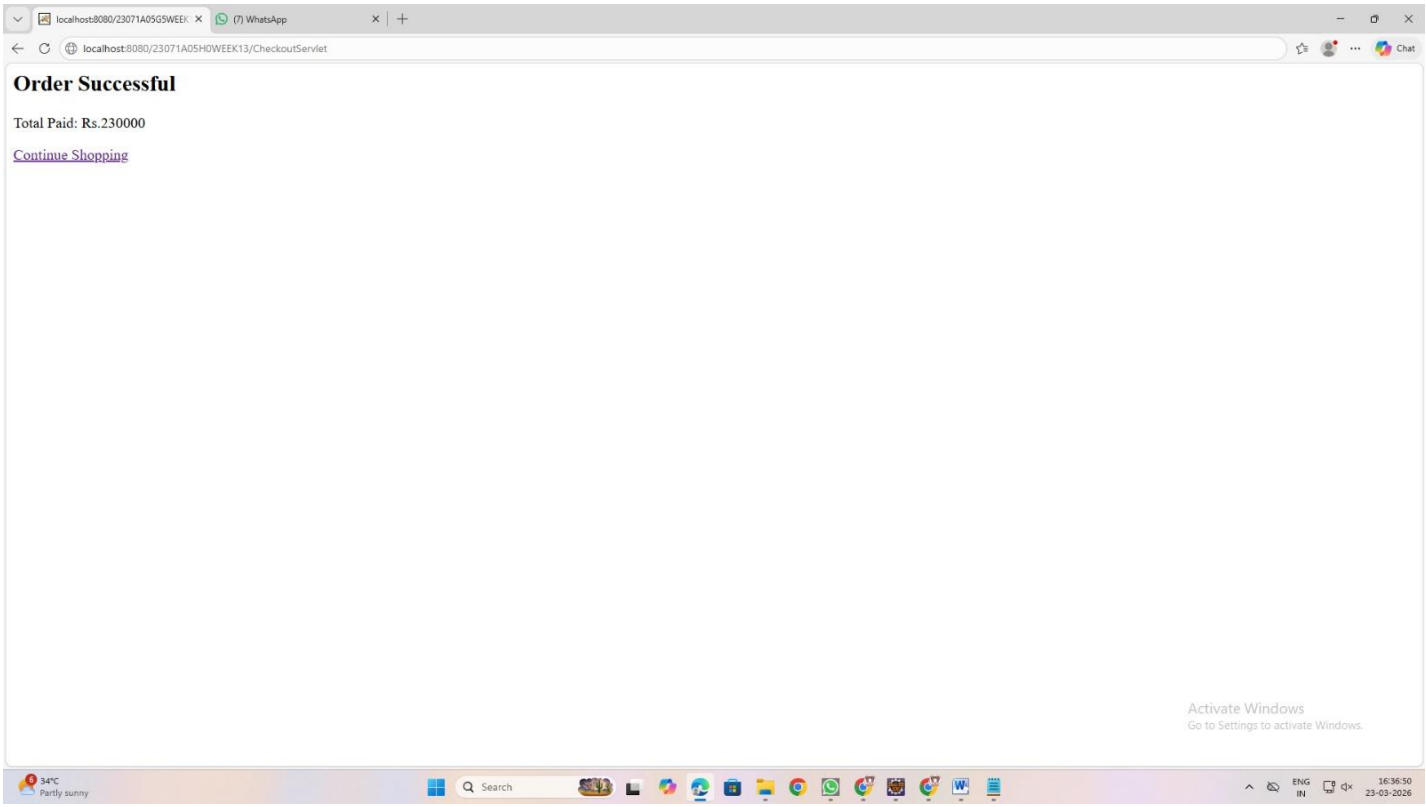
OUTPUT:


Register
localhost:8080/23071A05H0WEEK13/register.jsp
Name:
VAMSHI
Email:
23071A05H0@gmail.com
Password:
Register
Back to Login


Login
localhost:8080/23071A05G5WEEK13/login.jsp
Email:
23071A05H0@gmail.com
Password:
Login
Register


Catalog
(7) WhatsApp
localhost:8080/23071A05H0WEEK13/catalog.jsp
Product Catalog
Laptop
Price: 60000
Add to Cart
Mobile
Price: 25000
Add to Cart
Headphones
Price: 3000
Add to Cart
View Cart
Activate Windows
Go to Settings to activate Windows.


(7) WhatsApp
localhost:8080/23071A05G5WEEK13/checkout.jsp
Checkout
Laptop - Rs.180000
Mobile - Rs.50000
Total: Rs.230000
Name:
VAMSHI
Address:
ERRAGADDA , INDIAN BANK KALYANNAGAR BRANCH
Place Order
Activate Windows
Go to Settings to activate Windows.


localhost:8080/23071A05G5WEEK
(7) WhatsApp
localhost:8080/23071A05H0WEEK13/CheckoutServlet
Order Successful
Total Paid: Rs.230000
Continue Shopping
Activate Windows
Go to Settings to activate Windows.